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Date :- 19-8-26

Electronics and Communication Engineering

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- Objective
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Machine Learning

The Machine Learning Laboratory provides practical exposure to core machine learning concepts and algorithms used for data analysis, classification, prediction, and decision-making. The experiments cover supervised, unsupervised, and concept learning techniques, including neural networks, regression, clustering, probabilistic models, and decision trees. This laboratory helps students understand the practical implementation of machine learning algorithms and their applications in real-world problem-solving.

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To study and implement Hierarchical Clustering for data grouping

The aim of the experiment is to learn and perform the working of Hierarchical Clustering.

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To study and implement Hierarchical Clustering for data grouping

Introduction

Hierarchical clustering is an unsupervised machine learning algorithm used to group unlabeled datasets into clusters in a hierarchical manner. It is also known as hierarchical cluster analysis or HCA. This clustering method builds a tree-like structure, known as a dendrogram, to represent the relationships and hierarchy among the data points. Sometimes, the results of K-means clustering and hierarchical clustering may appear similar, but they differ based on their working mechanisms. Unlike the K-Means algorithm, there is no requirement to predetermine the number of clusters.

Basically, there are two types of hierarchical Clustering:

1. Agglomerative Clustering
2. Divisive Clustering

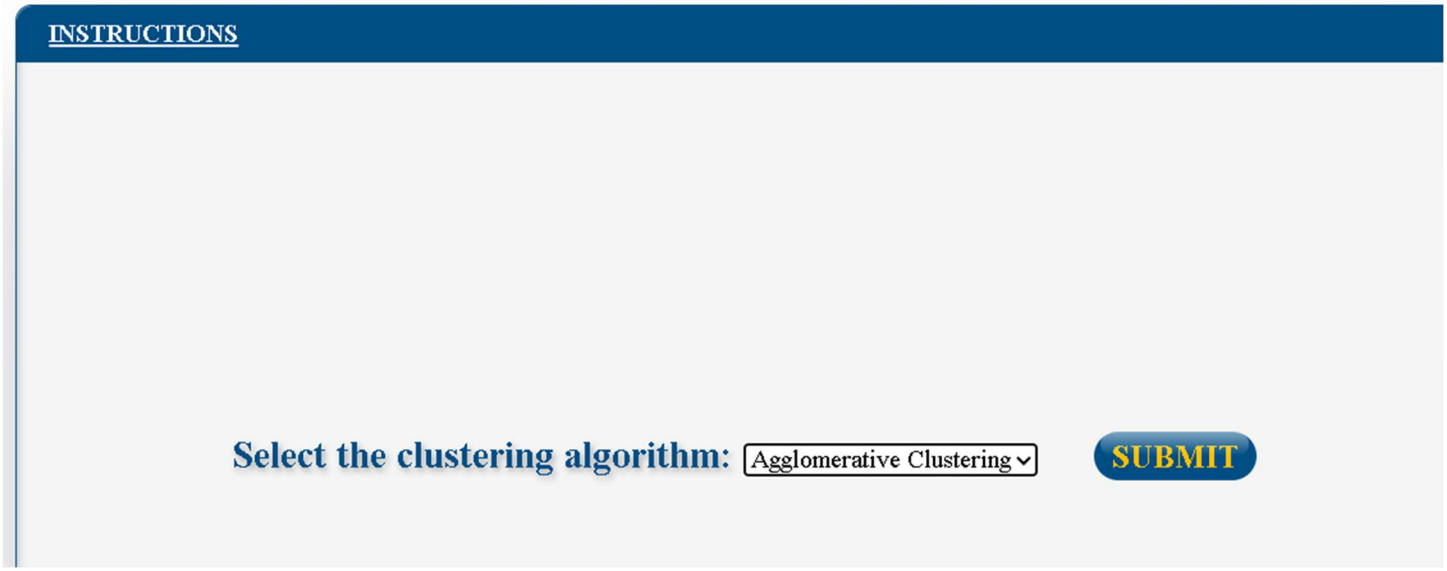
To study and implement Hierarchical Clustering for data grouping

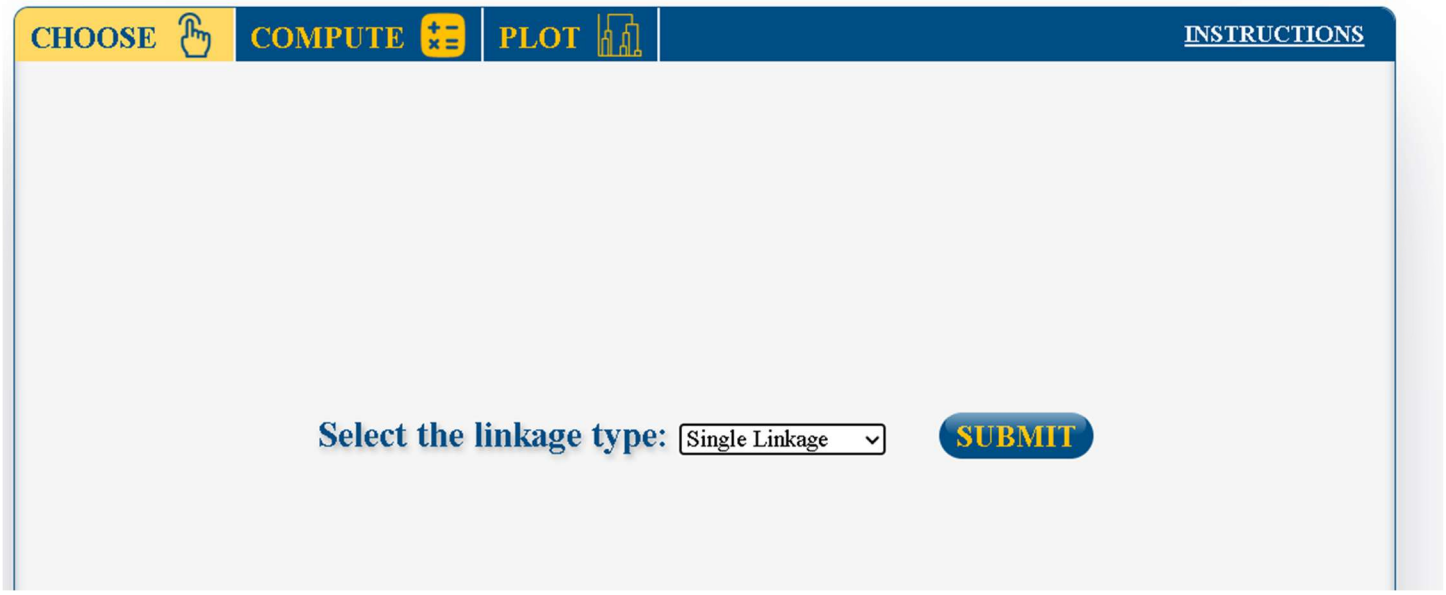
Agglomerative Clustering

Single Linkage

1. Calculate the Euclidean distance for the highlighted cell using the provided calculator (take X_1 & X_2 from the table). Next, click on the corresponding cell in the matrix and enter the calculated distance.
2. Repeat the step 1 until you've completed the matrix. (Mirrored values will be automatically filled)
3. Identify the two clusters with the smallest maximum pairwise distance by clicking on the MIN button.
4. Click the **MERGE** button to combine the two clusters from step 3 into a single cluster & update the matrix by recalculating pairwise distances, considering the single linkage criterion
5. Repeat step 3-4 until only one cluster remains.
6. Now, click on the **NEXT** button to plot the dendrogram.
7. Finally, click on the **PLOT** button.

Complete Linkage





DISTANCE MATRIX

	P1=18	P2=22	P3=25	P4=27	P5=42	P6=43
P1=18	0					
P2=22		0				
P3=25			0			
P4=27				0		
P5=42					0	
P6=43						0

CLUSTER:

FORMULA

Euclidean Distance:

$$\sqrt{(X_2 - X_1)^2}$$

Where, X_1 & X_2 are data points.

SINGLE LINKAGE

Single linkage clustering defines the distance between two clusters as the **minimum distance** between any pair of points belonging to different clusters.

How it works:

1. Each data point is initially treated as a separate cluster.
2. All pairwise distances between data points are computed.
3. The two closest points **(having minimum distance)** are identified and merged together to form a single cluster.
4. Distances between clusters are recalculated using the **single linkage** rule, where the minimum inter-point distance is considered.
5. The process is repeated until the stopping condition is satisfied or a single cluster remains.

CHOOSE COMPUTE PLOT INSTRUCTION

DISTANCE MATRIX

	P1=18	P2=22	P3=25	P4=27	P5=42	P6=43
P1=18	0	4	7	9	24	25
P2=22	4	0	3	5	20	21
P3=25	7	3	0	2	17	18
P4=27	9	5	2	0	15	16
P5=42	24	20	17	15	0	1
P6=43	25	21	18	16	1	0

CLUSTER:

FORMULA

Euclidean Distance:

$$\sqrt{(X_2 - X_1)^2}$$

Where, X_1 & X_2 are data points.

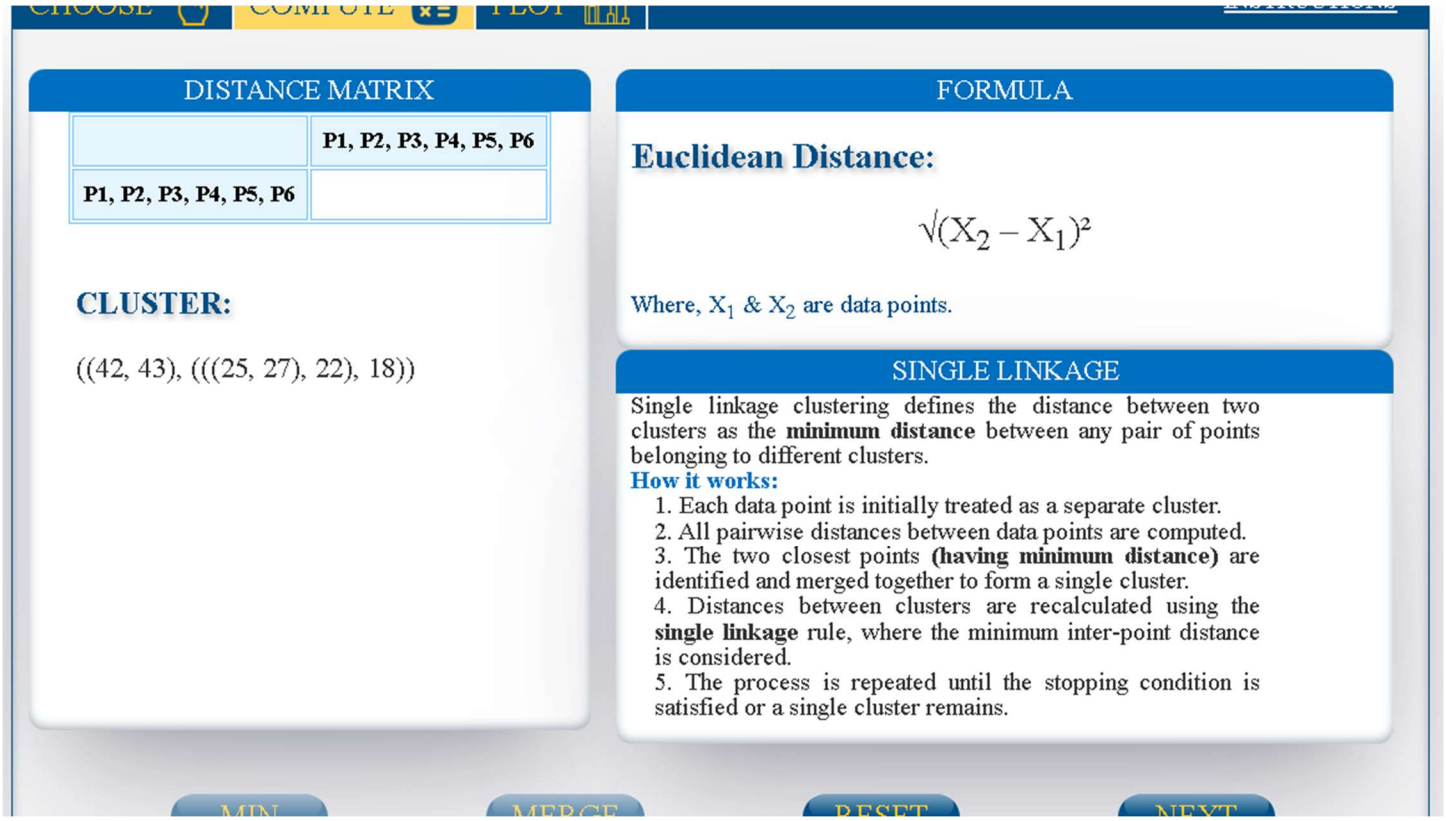
SINGLE LINKAGE

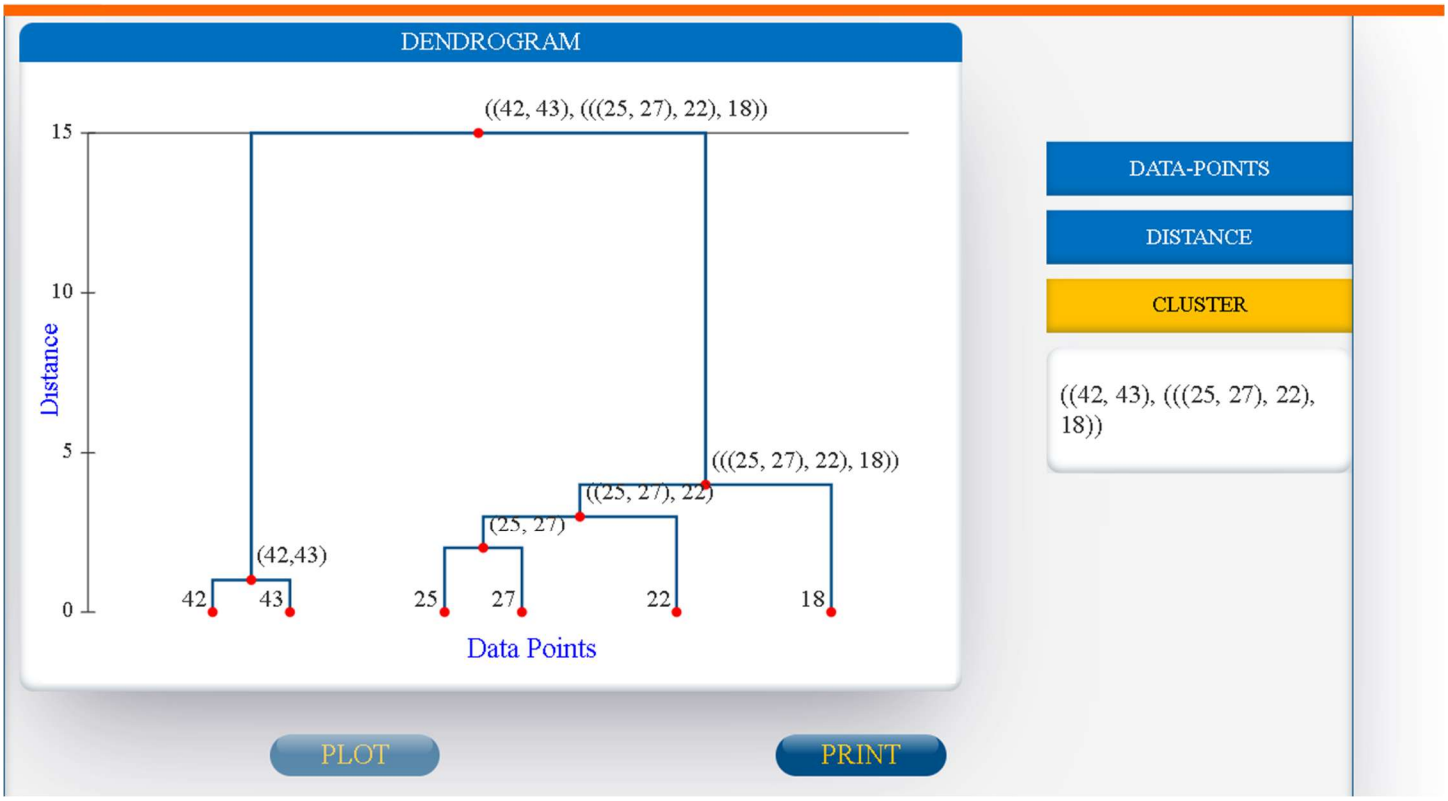
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5. The process is repeated until the stopping condition is satisfied or a single cluster remains.

MIN MERGE RESET NEXT





Conclusion:

Thus , we have experimented unsupervised machine learning techniques Hierarchical Clustering using virtual lab.